

# SPCE 5025

Class 3

3 Feb 2026

# Administrative

- Exam 1 next week
  - Exam will be distributed this week
  - Due Saturday, 14 Feb 2026

# Overview

- MATLAB moment
- Orbit State Representations
- Coordinate Systems
- Coordinate Transformations

# MATLAB: Calling a Function

```
fid = fopen('Z:\Matlab Stuff\keplerian.txt','wt');

mu = 3.986004418e14;

r = [326151.080726; 6077471.251787; 2944583.918767];
rd = [-7455.178720; -482.482572; 1910.883434];

fprintf(fid,'r: %16.6f %16.6f %16.6f meters\n', r(1), r(2), r(3));
fprintf(fid,'rd: %16.6f %16.6f %16.6f m/sec\n', rd(1), rd(2), rd(3));
fprintf(fid,'\n');

[a, e, inc, O, wp, nu, MA] = kepler(mu, r, rd);

fprintf(fid,'a: %16.8f m\n', a);
fprintf(fid,'e: %16.8f m\n', e);
fprintf(fid,'inc: %16.8f m\n', inc*180/pi);
fprintf(fid,'O: %16.8f m\n', O*180/pi);
fprintf(fid,'wp: %16.8f m\n', wp*180/pi);
fprintf(fid,'nu: %16.8f m\n', nu*180/pi);
fprintf(fid,'MA: %16.8f m\n', MA*180/pi);
fprintf(fid,'\n');
fprintf(fid,'\n');

fclose(fid);
```

# MATLAB: Calling a Function (2)

- Rules
  - Function should be its own file
    - Function name and MATLAB file name must be the same
  - Keep main and function in same folder
- Basic format of function:

```
function [ a, e, i, raan, wp, nu, MA ] = kepler( mu, r, rd)

%Introductory material and comments

                                ... CODE GOES HERE ...

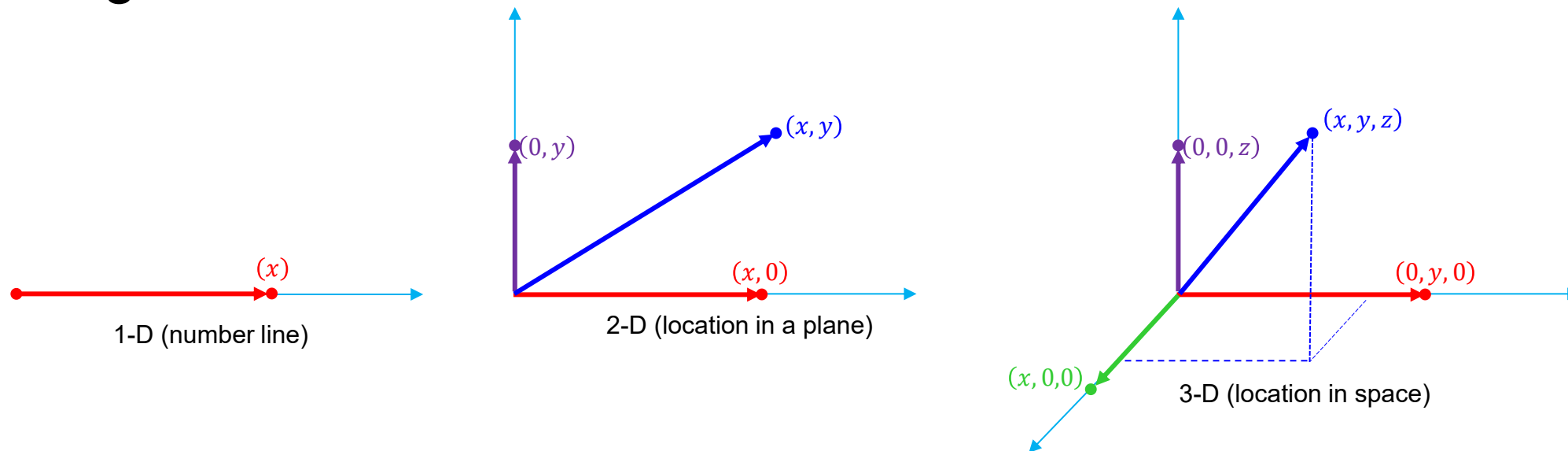
end
```

# Coordinate Systems

- **def** – Coordinate – a set of  $n$  variables that fix a geometric object
- **def** – Coordinate System – a system for specifying points using coordinates measured in some specified way.
- The simplest coordinate system consists of coordinate axes oriented perpendicularly to each other, known as Cartesian coordinates

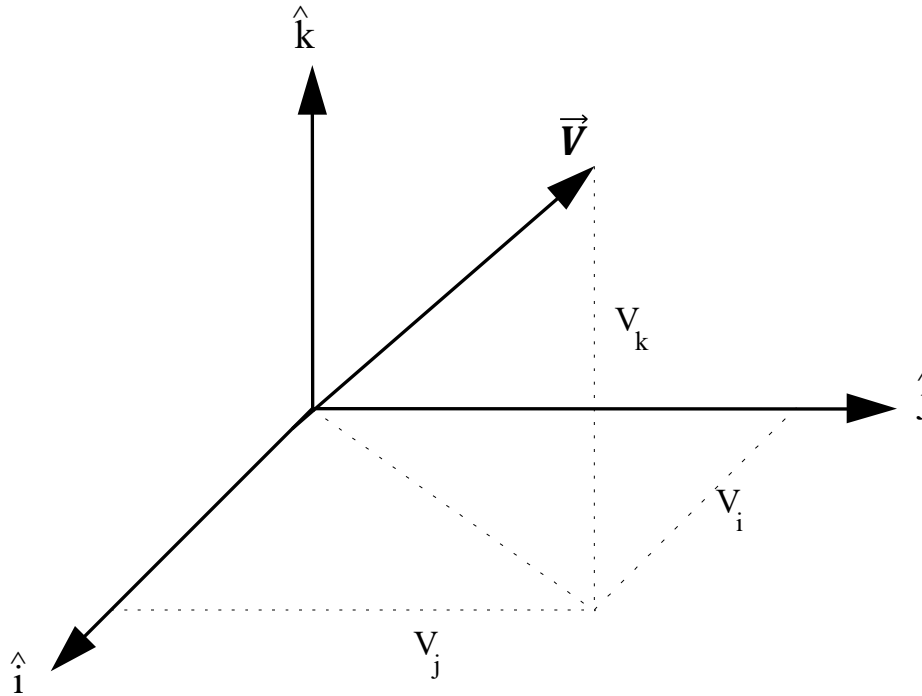
# Vector Representations

- **Vector** – an entity that has magnitude and direction.
- Measured with respect to some reference frame having a set of defining characteristics.
- The magnitude of a vector is defined in terms of distance from the origin



# Cartesian Representation

- Vector is represented as components, or distances, along each of the characteristic directions

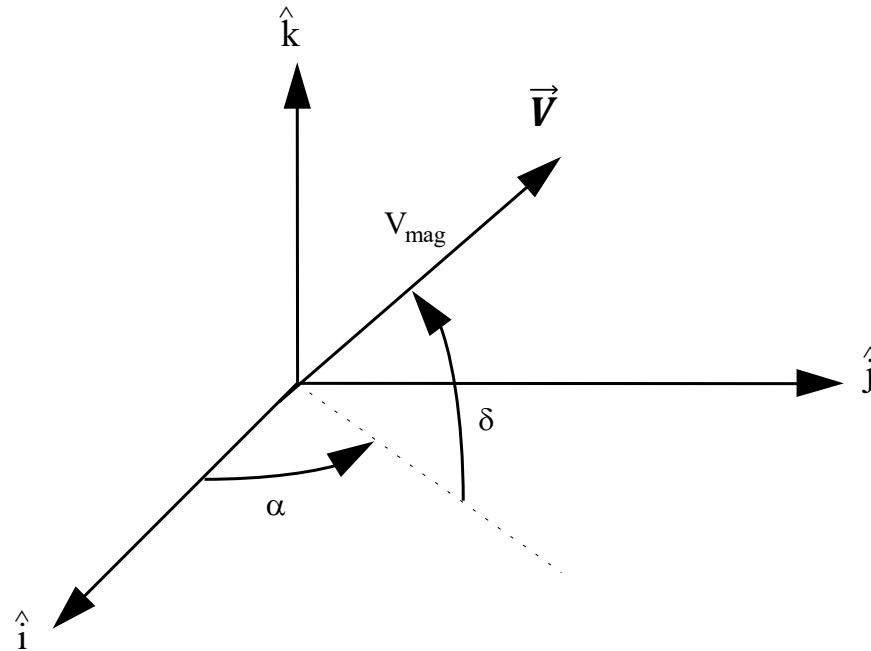


$$\vec{V} = v_i \hat{i} + v_j \hat{j} + v_k \hat{k}$$



# Right Ascension/Declination

- Vector is represented by magnitude and two angles
  - Usually right ascension  $\alpha$  and declination  $\delta$



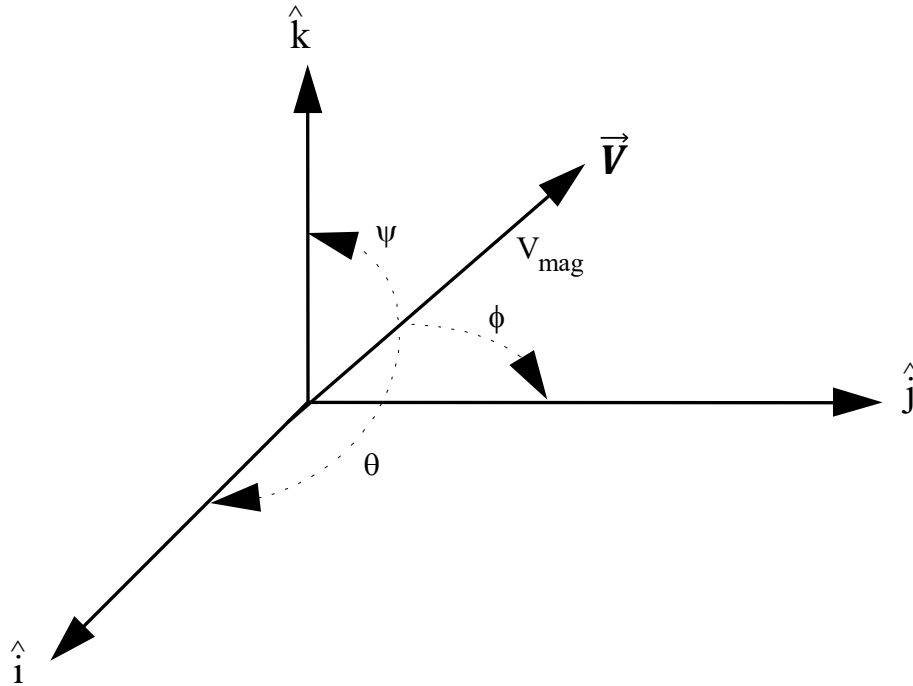
$$\begin{bmatrix} v_i \\ v_j \\ v_k \end{bmatrix} = v_{mag} \begin{bmatrix} \cos\alpha \cos\delta \\ \sin\alpha \cos\delta \\ \sin\delta \end{bmatrix}$$

- Compute  $\alpha$  and  $\delta$  from the vector components

$$\alpha = \text{ATAN2}(v_j, v_i) \qquad \delta = \sin^{-1} \left( \frac{v_k}{|\vec{V}|} \right)$$

# Direction Cosines

- Vector components related to angles between vector and each axis



$$\begin{bmatrix} V_i \\ V_j \\ V_k \end{bmatrix} = V_{\text{mag}} \begin{bmatrix} \cos\theta \\ \cos\phi \\ \cos\psi \end{bmatrix}$$

- Typically used for coordinate transformations

# Unit Vectors

- Unit-length vectors are convenient for use in coordinate system representations

$$\hat{\mathbf{v}} = \frac{\vec{\mathbf{v}}}{|\vec{\mathbf{v}}|} \qquad |\vec{\mathbf{v}}| = \sqrt{v_1^2 + v_2^2 + v_3^2}$$

- Some properties
  - Dimensionless
  - Unit length provides directionality without scaling effects
  - Example:

$$\ddot{\vec{\mathbf{r}}} = -\frac{\mu}{r^3} \vec{\mathbf{r}} = -\frac{\mu}{r^2} \hat{\mathbf{r}}$$

# Earth-Centered Coordinates

- Origin is at center of Earth
- Commonly used frames
  - Earth Centered Inertial (ECI) – many different types
    - J2000 (Inertial)
    - True of Date (Inertial)
    - Earth-Fixed Inertial (Instantaneous inertial)
  - Earth Centered-Earth-Fixed (Rotating)
- Earth-centered is natural frame for satellites orbiting the Earth

# Reference Frames

# Recommended Download

- See Chapter 3 of the Goddard Trajectory Determination System Math Spec
- Excellent, in-depth discussion of coordinate and time systems
- Available in Canvas content folder

# Inertial Reference Frames

- Reference frame defined by basis directions that are fixed with respect to immovable reference points
  - Allows integration of equations of motion without having to account for motion of coordinate axes
- Usually (notionally) based on celestial objects
  - Not immovable, but so far away that their apparent motion is negligible
  - We often pick their positions at a fixed epoch
- Common definitions for Earth-centered frames are based on:
  - Directions fixed at some epoch
  - Z axis oriented along Earth rotation axis (somehow)
  - X axis commonly in Equatorial plane through  $0^\circ$  longitude
  - Y axis completes the right-handed system

# Instantaneous/Quasi-Inertial

- Axes of coordinate system treated as if inertial, even though they're really not
- Axes are “frozen” at specified epoch
- Derived from axes that are not intrinsically fixed
  - Examples:
    - Instantaneous position of Earth-fixed axes
    - Orbit-relative (e.g., radial, in-track, cross-track)



# Motion of the Earth

- **Precession** of the Earth's equatorial plane
  - 26,000 year period
  - Caused by gravitational attraction of sun and moon on Earth's equatorial bulge
- **Nutation** – short-term (18.6 year) variation in precession rate
  - Major driver is precession of moon's line of nodes
- **Rotation** – the nearly constant rotation of the Earth about its axis
  - Small variations in rotation rate due to changes in mass distribution, etc.
- **Polar Motion** – small variations in direction of polar axis due to snow/water distribution, motions of Earth core and mantle, etc.